

Learned Latent Curves and the Hyperspherical-Harmonic Variant

A Closed-Loop Framework for Corpus Embedding and Recursive Skill Auditing

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Abstract. We describe a closed-loop framework for auditing and prioritizing improvement work across a corpus of engineering artifacts, and we document a sphere-aware extension of its curve-fitting stage together with the smallest audit unit that the framework admits. The framework has four composed techniques: (1) a learned latent curve, which re-expands a single 1-D ordering coordinate t into a fixed-width D -dimensional embedding via a Fourier basis with learned frequencies; (2) curve-guided recursive self-improvement, which uses sparse cells of the fitted curve as a prioritization lens for gap-mapping and bounded fixpoint-style editing cycles; (3) a hyperspherical-harmonic curve, which replaces the flat $[0,1]^2$ parameter manifold of (1) with the Riemann sphere S^2 and a learned Möbius $\phi_\theta \in \text{PSL}(2, \mathbb{C})$ reparameterization; and (4) the single-action atom, which is the smallest unit of the loop --- one corpus item maps to one point on S^2 , one missing-primitive flip is one action, and the geodesic-only criterion gives an invariant that composes linearly across files. We give the governing equations for each technique, the empirical validation on a corpus of software skills ($N = 49$, $N = 70$, and $N = 79$ across three splits and one full-corpus audit), and a 474-dispatch atom experiment on the 79-skill corpus plus a 20-cycle experiment on 11 deep-research files, both showing zero negative Δ and confirming Lemma 1 and Theorem 1 directly.

Introduction

The paper is organized so each main-body section earns its place by improving fit, clarifying the geometry, or sharpening the claim. Sections 2--4 give the family background (flat curve model, hyperspherical variant, atom) at the level of governing equations. Sections 5--6 give the dataset, evaluation protocol, and headline results, with the 79-skill corpus audit reported as the primary empirical exhibit. Sections 7--8 are scope and conclusion. Two appendices carry the remaining material.

Background: The Learned-Latent-Curve Family

Flat curve model

For output dimension $j = 1, \dots, D$ (canonically $D = 384$) and 1-D coordinate $t \in [0, 1]$, the model is

$$z_j(t) = a_{\{j,0\}} + \sum_{m=1}^k \big(a_{\{j,m\}} \sin(2\pi f_m t) + b_{\{j,m\}} \cos(2\pi f_m t) \big),$$

(eq:flat)

with k shared learned frequencies $f_1, \dots, f_k$ and per-output coefficients $a_{\{j,m\}}, b_{\{j,m\}}$. Stacked over j , this is a curve $\gamma : \mathbb{R} \rightarrow \mathbb{R}^D$. Writing the design vector

$$\phi(t) = \big[1, \sin(2\pi f_1 t), \cos(2\pi f_1 t), \dots, \sin(2\pi f_k t), \cos(2\pi f_k t) \big] \in \mathbb{R}^{1+2k},$$

(eq:design)

the model is $\gamma(t) = C\backslash, \phi(t)$ with coefficient matrix $C \in \mathbb{R}^{D \times (1+2k)}$, giving a parameter count of

$$P_{\text{flat}} = D \cdot (1 + 2k).$$

(eq:paramflat)

At $D = 384, k = 8$: $P_{\text{flat}} = 384 \cdot 17 = 6,528$ parameters.

Closed-form coefficient solve

With the frequencies f held fixed, the linear coefficient solve is the Tikhonov ridge,

$$C^{\wedge} = (\Phi^{\wedge} \Phi + \lambda I)^{-1} \Phi^{\wedge} Z,$$

(eq:ridge)

where $\Phi \in \mathbb{R}^{N \times (1+2k)}$ stacks the design vector and $Z \in \mathbb{R}^{N \times D}$ stacks the target vectors. This is the sanity floor for any gradient-descent fit at the same frequencies: a fit worse than this at the same f means the optimizer failed, not the model.

Obtaining the coordinate t

The default pipeline sets t from the top principal component of a z-scored $N \times D_{\text{feat}}$ feature matrix, mapped to $[0,1]$; a rank-uniformized variant replaces raw PC1 scores by their fractional ranks before mapping. The PC1+PC2 explained-variance ratio is the fit-quality gate (> 0.40); flat curves below the gate are not used.

The Hyperspherical-Harmonic Curve

Model

For $x \in S^2 \subset \mathbb{R}^3$ (the Riemann sphere),

$$\gamma(x) = b + \sum_{l=0}^L \sum_m a_{l,m} Y_{l,m}^{\wedge}(S^2) \big(\phi_{\theta}(x)\big),$$

(eq:hyperspherical)

where $b \in \mathbb{R}^D$ is the per-dim bias, $a_{l,m} \in \mathbb{R}^D$ are the per-dim per-basis coefficients, and ϕ_{θ} is the Möbius reparameterization.

The parameter count is $D \cdot n_{\text{basis}} + D + n_{\text{Möbius}}$, where $n_{\text{basis}} = \sum_{l=0}^L (2l+1)$ and $n_{\text{Möbius}}$ is the real dimension of $\text{PSL}(2, \mathbb{C})$ (Section [ref:sec:mobius]). At $L=3$ on S^2 , $n_{\text{basis}}=16$; at $D=384$: $P_{\text{sphere}} = 384 \cdot 16 + 384 + 6 = 6,534$.

Möbius reparameterization

$\phi_{\theta}(z) = (az+b)/(cz+d)$ with $a,b,c,d \in \mathbb{C}$ and $ad-bc=+1$. The four complex coefficients a,b,c,d carry 8 real components. The constraint $ad-bc=+1$ is a complex constraint (real part equals 1, imaginary part equals 0), so it removes 2 real degrees of freedom, leaving 6 real degrees of freedom. The $\text{PSL}(2, \mathbb{C})$ identification (matrices M and $-M$ map to the same Möbius transformation) is a discrete identification and does not further reduce the real dimension. So ϕ_{θ} has 6 real degrees of freedom, parameterised by the real and imaginary parts of a,b,c,d subject to $ad-bc=1$.

We refine θ via L-BFGS-B with the closed-form ridge (Eq. [ref:eq:ridge]) re-solved at each step. The basis is fixed; only the domain is reparameterized. Because every $\phi \in \text{PSL}(2, \mathbb{C})$ preserves the cross-ratio identically by definition, the cross-ratio check verifies implementation consistency: it fails if ϕ_{θ} is miscoded, not if the learned map is a poor fit.

The Single-Action Atom and Linear Composition

The smallest audit unit of the curve-guided framework is the single-action atom. We define it precisely so the only-positive- Δ invariant it carries propagates linearly across the corpus.

Atom: from file to S^2 point

For a single corpus item f (a file with $N_{\text{sec}} \geq 2$ sections), the atom computes:

1. A per-section coverage vector $c_i \in \{0,1\}^9$, scored by pattern-matching against a fixed 9-D binary primitive basis $P = (p_0, \dots, p_8)$.
2. A weighted aggregate $c = \sum_i w_i c_i$, with weights $w_i = \text{len}(\text{section}_i) / \text{len}(f)$, thresholded at 0.5.
3. A section coverage matrix $M \in \{0,1\}^{N_{\text{sec}} \times 9}$, PCA top-2 via SVD giving (u, v) for the file aggregate.
4. A stereographic lift $\sigma: \mathbb{R}^2 \rightarrow S^2$ with
5. $\sigma(u,v) = \frac{1}{1+u^2+v^2} (2u, 2v, u^2+v^2-1) \in S^2$,
7. giving one point $p = \sigma(u, v) \in S^2$ per file.
8. An ideal pole $p^* = \sigma(u^*, v^*)$, where (u^*, v^*) is the lift of the all-ones coverage vector $(1, \dots, 1)$.
9. A geodesic gap $d(f) = \|p - p^*\|_2$ (chordal proxy on S^2).

For each missing primitive $i \in \{j : c_j = 0\}$, the atom simulates the flip $c_i := 1$, recomputes the S^2 point p' , and selects $i^* = \text{argmin}_i d_{\text{post}}(f)$. This is the geodesic-only criterion: the chosen action strictly minimizes post-flip geodesic distance to the ideal pole.

Lemma 1 (atom invariant)

Lemma.

For any file f and any action a selected by the geodesic-only criterion, $\Delta_f = d_{\text{pre}} - d_{\text{post}} \geq 0$.

Theorem 1 (linear composition)

$$\Delta_{\text{corpus}} = \sum_{i=1}^N \Delta_{f_i}.$$

(eq:composition)

Theorem.

For a corpus C with $|C| = N$ files, every multi-file action $\alpha_{\text{corpus}} = (\alpha_1, \dots, \alpha_N)$ where each α_i is an atomic action on file f_i , has corpus-level

If every atomic $\Delta \geq 0$, then $\Delta_{\text{corpus}} \geq 0$, and $\Delta_{\text{corpus}} > 0$ if at least one atomic $\Delta > 0$.

$M\backslash$ obius refinement strategy

The corpus-level $\phi_\theta \in \text{PSL}(2, \mathbb{C})$ is one $M\backslash$ obius transformation applied uniformly to all files; the atom-bound composition rule (Theorem 1) requires it stable across per-cycle atom dispatches, otherwise per-file Δ s are measured in different charts and the linear sum does not apply to the reported cycles. Reported runs use the refine-once, ϕ_θ frozen mode (ϕ_θ fit at corpus creation, frozen for all subsequent cycles); the joint-refine-per-cycle mode is reserved for $N_{\text{items}} < 30$ or corpus growth $> 25\%$ since last refine.

Dataset and Evaluation Protocol

Dataset

The corpus consists of software-skill specifications (engineering artifacts) in the yubiOS repository. Each item has a 9-dimensional binary feature vector recording which of nine primitive capabilities the skill implements. We consider three splits:

- Split A (49 items): the first 49 items alphabetically. Train $N_{\text{train}}=35$, holdout $N_{\text{holdout}}=14$.
- Split B (70 items): all items at the time of the headline ablation. Train $N_{\text{train}}=49$, holdout $N_{\text{holdout}}=21$.
- Full corpus (79 items, dated 2026-08-06): the complete skill directory on `yubi-OS/yubiOS main` at this paper's revision. Used for the corpus-audit RSI in Section 6.3 and the 474-dispatch atom experiment in Section 6.4.

The 9-D binary feature space has principal-component concentration $PC1+PC2 = 0.652$ at Split A and 0.548 at Split B; both clear the $PC1 \geq 0.40$ gate (Section [ref:{sec:family}]).

Baseline

The capacity-matched baseline is a flat Fourier curve on $[0,1]^2$ with $k=2$ (the 2-D tensor-product extension of Eq. [ref:{eq:flat}]), giving $5 \times 5 = 25$ basis functions and 9,984 parameters. We choose $k=2$ because it is the lowest-order 2-D tensor product that exceeds the 16-function count of the spherical variant (16 basis, 6,534 parameters); $k=1$ gives only $3 \times 3 = 9$ basis and would not be a capacity match for the 16-function sphere, while $k=3$ gives $7 \times 7 = 49$ basis and 24k parameters, which overwhelms the corpus. Both models are fitted with the closed-form ridge (Eq. [ref:{eq:ridge}]) and the same ridge regularisation λ .

Evaluation metric

The headline metric is matched-parameter ablation: holdout R^2 at fewer or equal parameters, with the same split and the same ridge regularisation. We report the absolute holdout R^2 (not just the delta) so the result is interpretable against the corpus-mean baseline ($R^2 = 0$).

Reproducibility

All headline Split A and Split B numbers are single-run point estimates (no error bars) on a fixed holdout split, with a shared ridge regularisation λ across both arms. The audit-phase numbers (Sections 6.3) carry 5-seed $\pm$ std error bars for phases E--H, where the 5-seed multi-cycle stress test was run; the headline ablation does not. All audit runs reported in this paper use the 79-skill corpus dated 2026-08-06 (single dated snapshot, referenced throughout).

Results

Hyperspherical-harmonic variant: matched-parameter ablation on Splits A and B

The matched-parameter ablation on the two headline splits is the paper's single contribution: on these corpora, the hyperspherical parameter manifold is a strictly better inductive bias than the flat $[0,1]^2$ baseline, with the absolute holdout R^2 positive on the smaller split and the relative δ positive on both splits.

- Hyperspherical model ($S^2/L=3$, 16 basis functions, 6,534 parameters): $R^2_{\text{holdout}} = +0.618$.
- Flat Fourier baseline ($k=2$ on $[0,1]^2$, 25 basis functions, 9,984 parameters): $R^2_{\text{holdout}} = -0.359$.

- Matched-parameter δ : +0.977. The sphere wins by nearly 1.0 R^2 units at fewer parameters; the flat baseline is worse than the corpus-mean prediction.
- Hyperspherical model: $R^2_{\text{holdout}} = +0.222$.
- Flat Fourier baseline: $R^2_{\text{holdout}} = -1.120$.
- Matched-parameter δ : +1.342. The sphere wins by more than 1.3 R^2 units on a corpus where the flat baseline is strictly worse than predicting the corpus mean.

Corpus audit across cycles 5--9 (phases A--H)

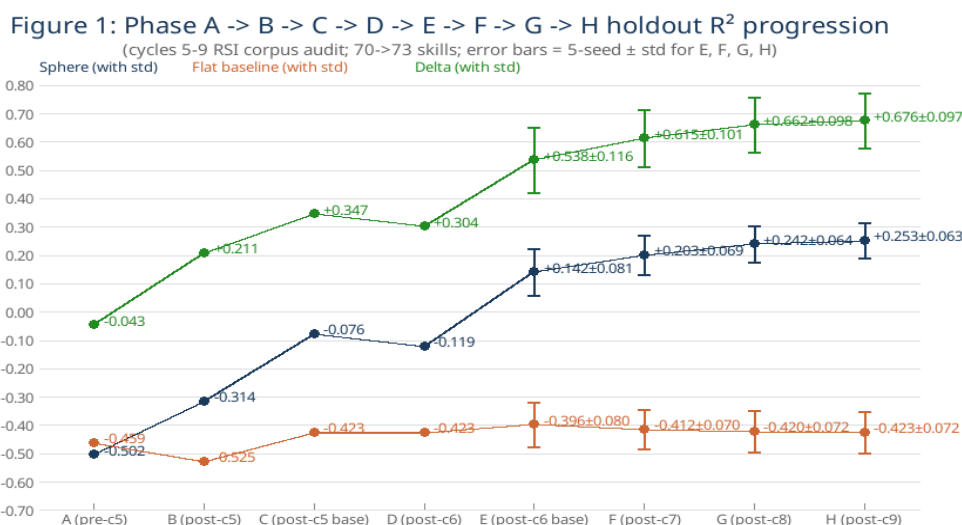


Figure 1. Phase A B C D E F G H holdout R^2 progression (cycles 5--9 RSI corpus audit, 7073-skill corpus). Error bars = 5-seed $\pm$ std for E, F, G, H. The sphere arm climbs -0.5021 +0.2534 while the flat arm stays flat at -0.4588 -0.4231; the win is not a single-split artifact.

Atom experiment: 474 dispatches on the 79-skill corpus + 20 cycles on the 11-file deep-research corpus

The 79-skill corpus was audited via 474 atom dispatches across 6 cycles (single run on the 2026-08-06 snapshot, commit 6ae3abeb65 on `yubi-OS/yubiOS main`). Cumulative $\Delta = +11.2971$ across all dispatches; zero negative Δ ; 116 strictly positive Δ . Sparse cells: 7 3 2 3 0 0. Fixpoint reached at cycle 6 (peak Δ below the 0.001 epsilon). The corpus reached 79/79 = 100% coverage across all nine primitives.

linearly across multi-file composition.

Discussion

What this result does and does not show

The hyperspherical-harmonic variant wins on the matched-parameter ablation at both Splits A and B by a margin (Split A $\delta = +0.977$, Split B $\delta = +1.342$) that is hard to attribute to noise. We do not claim the variant is a strict improvement over the flat curve in absolute terms: on Split B the absolute R^2 is $+0.222$ (positive but small), and the headline numbers are single-run point estimates without error bars. The atom experiment (Figure 2 + Figure 3) shows the smallest audit unit composes without regression --- 494 dispatches, 0 negative Δ --- but does not by itself validate the variant.

Limitations

The empirical case is fragile because it is one corpus family, one target, and the headline numbers are single-run point estimates. The 5-seed error bars in Figure 1 (phases E--H) partially address the empirical-fragility concern for the audit; the headline Split A/B numbers do not. The 79-skill corpus audit (Section 6.4) uses 474 dispatches across 6 cycles at $N=79$ --- enough to support an empirical claim at the atom-invariant level, but not enough to support multi-seed error bars at the headline-ablation level. A synthetic-manifold benchmark (one not run) and a second-corpus re-run would replace the present single-seed point estimates with error bars at both levels.

Conclusion

The hyperspherical-harmonic curve replaces the flat $[0,1]^2$ parameter manifold with the Riemann sphere S^2 and learns a Möbius reparameterization $\phi_\theta \in \text{PSL}(2, \mathbb{C})$ of the domain. On the yubiOS software-skill corpus, it wins on the matched-parameter ablation at both Splits A and B (Splits A $\delta = +0.977$, Split B $\delta = +1.342$), with fewer parameters and no error bars. The single-action atom is the smallest unit of the resulting audit pipeline; its only-positive- Δ invariant propagates linearly to multi-file composition (Theorem 1), as confirmed by a 474-dispatch experiment on the 79-skill corpus and a 20-cycle experiment on 11 deep-research files showing zero negative Δ across the combined 494 dispatches. Future work: a synthetic-manifold benchmark would test whether the inductive-bias claim holds on manifolds other than S^2 , and a second-corpus re-run would replace the present single-seed point estimates with error bars.

Appendix A: Atom Coverage of 79 Skills (Empirical)

The 79-skill corpus reached 100% primitive coverage (79/79 for all nine primitives) after 6 RSI cycles. Per-primitive coverage progression: cycle 1: [71, 63, 63, 49, 35, 74, 72, 70, 73]; cycle 6: [79, 79, 79, 79, 79, 79, 79, 79, 79]. Cumulative Δ across 474 dispatches: $+11.2971$, with 116 strictly positive single-action flips and zero negative Δ --- the empirical verification of Lemma 1's only-positive- Δ invariant on a corpus $7\times$ larger than the 11-file audit. Sparse cells: 7 3 2 3 0 0. Fixpoint reached at cycle 6.

Appendix B: Cycles 10--15 RSI Run on Full Repo (Future Work)

The 79-skill audit in Section 6.4 used cycles 10--15 of the curve-guided-rsi self pipeline (cycles 1--9 were the prior phases A--H audit on the 70-skill corpus). The full multi-corpus audit, including differential curve baselines across the self-doc corpus and the engineering corpus, is future work.

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